

Creation Date 20-Jul-2010

Revision Date 09-Feb-2024

Revision Number 5

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

**Product Description:** Tetrazole, 3 to 4 wt% solution in acetonitrile  
**Cat No. :** 432990000; 432990010; 432991000  
**Molecular Formula** C H<sub>2</sub> N<sub>4</sub>

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                       |                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Recommended Use</b>                | Laboratory chemicals.                                                                       |
| <b>Sector of use</b>                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| <b>Product category</b>               | PC21 - Laboratory chemicals                                                                 |
| <b>Process categories</b>             | PROC15 - Use as a laboratory reagent                                                        |
| <b>Environmental release category</b> | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| <b>Uses advised against</b>           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

##### EU entity/business name

Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

##### UK entity/business name

Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

##### Swiss distributor - Fisher Scientific AG

Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

#### E-mail address

begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:

Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

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## 2.1. Classification of the substance or mixture

### CLP Classification - Regulation (EC) No 1272/2008

#### Physical hazards

Flammable liquids

Category 2 (H225)

#### Health hazards

Acute oral toxicity

Category 4 (H302)

Acute dermal toxicity

Category 4 (H312)

Acute Inhalation Toxicity - Vapors

Category 4 (H332)

Serious Eye Damage/Eye Irritation

Category 2 (H319)

#### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### **Hazard Statements**

H225 - Highly flammable liquid and vapor

H319 - Causes serious eye irritation

H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled

### **Precautionary Statements**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P312 - Call a POISON CENTER or doctor if you feel unwell

P264 - Wash face, hands and any exposed skin thoroughly after handling

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P337 + P313 - If eye irritation persists: Get medical advice/attention

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

## 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Toxicity to Soil Dwelling Organisms

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

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## 3.2. Mixtures

| Component    | CAS No   | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                               |
|--------------|----------|-------------------|----------|-----------------------------------------------------------------------------------------------------------------|
| Acetonitrile | 75-05-8  | 200-835-2         | 96-97    | Flam. Liq. 2 (H225)<br>Acute Tox. 4 (H302)<br>Acute Tox. 4 (H312)<br>Eye Irrit. 2 (H319)<br>Acute Tox. 4 (H332) |
| 1H-Tetrazole | 288-94-8 | EEC No. 206-023-4 | 3-4      | Expl. 1.1 (H201)                                                                                                |

| Component    | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|--------------|-----------------------|-------------------------|-----------------------------|
| Acetonitrile | ATE = 617 mg/kg       | -                       | -                           |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | If symptoms persist, call a physician.                                                                                                           |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| Ingestion                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                                                 |
|--------------------|-------------------------------------------------|
| Notes to Physician | Treat symptomatically. Symptoms may be delayed. |
|--------------------|-------------------------------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

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Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

## Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>), Hydrogen cyanide (hydrocyanic acid).

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Ensure adequate ventilation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area. Keep away from heat, sparks and flame. Store under an inert atmosphere. Protect from moisture.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Class 3

**Switzerland - Storage of hazardous substances**

Storage class - SC 3  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

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## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component    | European Union                                               | The United Kingdom                                                                                                  | France                                                                                                                                                                 | Belgium                                                        | Spain                                                                                  |
|--------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Acetonitrile | TWA: 40 ppm (8hr)<br>TWA: 70 mg/m <sup>3</sup> (8hr)<br>Skin | STEL: 60 ppm 15 min<br>STEL: 102 mg/m <sup>3</sup> 15 min<br><br>TWA: 40 ppm 8 hr<br>TWA: 68 mg/m <sup>3</sup> 8 hr | TWA / VME: 40 ppm (8 heures). restrictive limit<br>TWA / VME: 70 mg/m <sup>3</sup> (8 heures). restrictive limit<br>TWA / VME: 5 mg/m <sup>3</sup> (8 heures).<br>Peau | TWA: 20 ppm 8 uren<br>TWA: 34 mg/m <sup>3</sup> 8 uren<br>Huid | TWA / VLA-ED: 40 ppm (8 horas)<br>TWA / VLA-ED: 68 mg/m <sup>3</sup> (8 horas)<br>Piel |

| Component    | Italy                                                                                                             | Germany                                                                                                                                                                                                                                                                                                                                         | Portugal                                                         | The Netherlands                  | Finland                                                                                                                                           |
|--------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetonitrile | TWA: 20 ppm 8 ore.<br>Time Weighted Average<br>TWA: 35 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>Pelle | TWA: 10 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 17 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 10 ppm (8 Stunden). MAK<br>TWA: 17 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 20 ppm<br>Höhepunkt: 34 mg/m <sup>3</sup><br>Höhepunkt: 2 mg/m <sup>3</sup><br>Haut | TWA: 40 ppm 8 horas<br>TWA: 70 mg/m <sup>3</sup> 8 horas<br>Pele | TWA: 34 mg/m <sup>3</sup> 8 uren | TWA: 20 ppm 8 tunteina<br>TWA: 34 mg/m <sup>3</sup> 8 tunteina<br>STEL: 40 ppm 15 minuutteina<br>STEL: 68 mg/m <sup>3</sup> 15 minuutteina<br>Iho |

| Component    | Austria                                                                                                                                                    | Denmark                                                                                                                                | Switzerland                                                                                                                                   | Poland                                                                           | Norway                                                                                                                                                                                                        |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetonitrile | Haut<br>MAK-KZGW: 160 ppm 15 Minuten<br>MAK-KZGW: 280 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 40 ppm 8 Stunden<br>MAK-TMW: 70 mg/m <sup>3</sup> 8 Stunden | TWA: 40 ppm 8 timer<br>TWA: 70 mg/m <sup>3</sup> 8 timer<br>STEL: 80 ppm 15 minutter<br>STEL: 140 mg/m <sup>3</sup> 15 minutter<br>Hud | Haut/Peau<br>STEL: 40 ppm 15 Minuten<br>STEL: 68 mg/m <sup>3</sup> 15 Minuten<br>TWA: 20 ppm 8 Stunden<br>TWA: 34 mg/m <sup>3</sup> 8 Stunden | STEL: 140 mg/m <sup>3</sup> 15 minutach<br>TWA: 70 mg/m <sup>3</sup> 8 godzinach | TWA: 30 ppm 8 timer<br>TWA: 50 mg/m <sup>3</sup> 8 timer<br>TWA: 5 mg/m <sup>3</sup> 8 timer<br>STEL: 45 ppm 15 minutter. value calculated<br>STEL: 75 mg/m <sup>3</sup> 15 minutter. value calculated<br>Hud |

| Component    | Bulgaria                                                  | Croatia                                                                      | Ireland                                                                                                                    | Cyprus                                   | Czech Republic                                                                                                |
|--------------|-----------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Acetonitrile | TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup><br>Skin notation | kože<br>TWA-GVI: 40 ppm 8 satima.<br>TWA-GVI: 70 mg/m <sup>3</sup> 8 satima. | TWA: 40 ppm 8 hr.<br>TWA: 70 mg/m <sup>3</sup> 8 hr.<br>STEL: 120 ppm 15 min<br>STEL: 310 mg/m <sup>3</sup> 15 min<br>Skin | TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> | TWA: 70 mg/m <sup>3</sup> 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 100 mg/m <sup>3</sup> |

| Component    | Estonia                                                                  | Gibraltar                                                           | Greece                                                                                  | Hungary                                                                          | Iceland                                                                                                                                          |
|--------------|--------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetonitrile | Nahk<br>TWA: 40 ppm 8 tundides.<br>TWA: 70 mg/m <sup>3</sup> 8 tundides. | Skin notation<br>TWA: 40 ppm 8 hr<br>TWA: 70 mg/m <sup>3</sup> 8 hr | STEL: 60 ppm<br>STEL: 105 mg/m <sup>3</sup><br>TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> | TWA: 70 mg/m <sup>3</sup> 8 órában. AK<br>lehetséges borón keresztül felszívódás | TWA: 40 ppm 8 klukkustundum.<br>TWA: 70 mg/m <sup>3</sup> 8 klukkustundum.<br>Skin notation<br>Ceiling: 80 ppm<br>Ceiling: 140 mg/m <sup>3</sup> |

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| Component    | Latvia                                                                              | Lithuania                                                 | Luxembourg                                                                                                         | Malta                                                                                          | Romania                                                               |
|--------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Acetonitrile | skin - potential for cutaneous exposure<br>TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> | TWA: 40 ppm IPRD<br>TWA: 70 mg/m <sup>3</sup> IPRD<br>Oda | Possibility of significant uptake through the skin<br>TWA: 40 ppm 8 Stunden<br>TWA: 70 mg/m <sup>3</sup> 8 Stunden | possibility of significant uptake through the skin<br>TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> | Skin notation<br>TWA: 40 ppm 8 ore<br>TWA: 70 mg/m <sup>3</sup> 8 ore |

| Component    | Russia                    | Slovak Republic                                                                | Slovenia                                                                                                                            | Sweden                                                                                                                                                                    | Turkey                                                         |
|--------------|---------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Acetonitrile | MAC: 10 mg/m <sup>3</sup> | Potential for cutaneous absorption<br>TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> | TWA: 40 ppm 8 urah<br>TWA: 70 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 140 mg/m <sup>3</sup> 15 minutah<br>STEL: 80 ppm 15 minutah | Indicative STEL: 60 ppm 15 minuter<br>Indicative STEL: 100 mg/m <sup>3</sup> 15 minuter<br>TLV: 30 ppm 8 timmar.<br>NGV<br>TLV: 50 mg/m <sup>3</sup> 8 timmar. NGV<br>Hud | Deri<br>TWA: 40 ppm 8 saat<br>TWA: 70 mg/m <sup>3</sup> 8 saat |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                         | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|-----------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Acetonitrile<br>75-05-8 ( 96-97 ) |                              |                                 |                                | DNEL = 32.2mg/kg bw/day           |

| Component                         | Acute effects local (Inhalation)        | Acute effects systemic (Inhalation)     | Chronic effects local (Inhalation)      | Chronic effects systemic (Inhalation)   |
|-----------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|
| Acetonitrile<br>75-05-8 ( 96-97 ) | DNEL = 40.6 ppm (68 mg/m <sup>3</sup> ) | DNEL = 40.6 ppm (68 mg/m <sup>3</sup> ) | DNEL = 40.6 ppm (68 mg/m <sup>3</sup> ) | DNEL = 40.6 ppm (68 mg/m <sup>3</sup> ) |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                         | Fresh water   | Fresh water sediment         | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|-----------------------------------|---------------|------------------------------|--------------------|------------------------------------|--------------------------|
| Acetonitrile<br>75-05-8 ( 96-97 ) | PNEC = 10mg/L | PNEC = 7.53mg/kg sediment dw | PNEC = 10mg/L      | PNEC = 32mg/L                      | PNEC = 2.41mg/kg soil dw |

| Component                         | Marine water | Marine water sediment | Marine water Intermittent | Food chain | Air |
|-----------------------------------|--------------|-----------------------|---------------------------|------------|-----|
| Acetonitrile<br>75-05-8 ( 96-97 ) | PNEC = 1mg/L |                       |                           |            |     |

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## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

Physical State Liquid

#### Appearance

##### Odor

No information available

##### Odor Threshold

No data available

##### Melting Point/Range

No data available

##### Softening Point

No data available

##### Boiling Point/Range

77 °C / 170.6 °F

##### Flammability (liquid)

Highly flammable

On basis of test data

##### Flammability (solid,gas)

Not applicable

Liquid

##### Explosion Limits

No data available

##### Flash Point

5 °C / 41 °F

**Method -** No information available

##### Autoignition Temperature

No data available

##### Decomposition Temperature

No data available

##### pH

No information available

##### Viscosity

No data available

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|                                         |                          |             |
|-----------------------------------------|--------------------------|-------------|
| Water Solubility                        | No information available |             |
| Solubility in other solvents            | No information available |             |
| Partition Coefficient (n-octanol/water) |                          |             |
| Component                               | log Pow                  |             |
| Acetonitrile                            | -0.34                    |             |
| Vapor Pressure                          | No data available        |             |
| Density / Specific Gravity              | 0.800                    |             |
| Bulk Density                            | Not applicable           | Liquid      |
| Vapor Density                           | No data available        | (Air = 1.0) |
| Particle characteristics                | Not applicable (liquid)  |             |

## 9.2. Other information

|                      |                                             |
|----------------------|---------------------------------------------|
| Molecular Formula    | C H2 N4                                     |
| Molecular Weight     | 70.05                                       |
| Explosive Properties | Vapors may form explosive mixtures with air |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

Yes

### 10.2. Chemical stability

Hygroscopic.

### 10.3. Possibility of hazardous reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

### 10.4. Conditions to avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moist air or water. Do not distill or allow to evaporate. Do not allow evaporation to dryness.

### 10.5. Incompatible materials

Strong oxidizing agents. Reducing Agent. Bases. Strong acids. Metals. Acid anhydrides. Acid chlorides.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Hydrogen cyanide (hydrocyanic acid).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

|            |            |
|------------|------------|
| Oral       | Category 4 |
| Dermal     | Category 4 |
| Inhalation | Category 4 |

#### Toxicology data for the components

| Component    | LD50 Oral                                 | LD50 Dermal             | LC50 Inhalation                                                             |
|--------------|-------------------------------------------|-------------------------|-----------------------------------------------------------------------------|
| Acetonitrile | 450-787 mg/kg (Rat)<br>2460 mg/kg ( Rat ) | > 2000 mg/kg ( Rabbit ) | LC50 = 3587 ppm (6.022 mg/l)<br>(Mouse) 4h<br>LC50 = 16,000 ppm (26.8 mg/l) |

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|              |                       |                         | (Rat) 4h                    |
|--------------|-----------------------|-------------------------|-----------------------------|
| Component    | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
| Acetonitrile | ATE = 617 mg/kg       | -                       | -                           |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory  
Skin

Based on available data, the classification criteria are not met

Based on available data, the classification criteria are not met

(e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

(f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

(h) STOT-single exposure;

Based on available data, the classification criteria are not met

(i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

Target Organs

None known.

(j) aspiration hazard;

Based on available data, the classification criteria are not met

Other Adverse Effects

The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## 11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects

Do not empty into drains. .

| Component    | Freshwater Fish                                                                                                                                                                                                                     | Water Flea | Freshwater Algae |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|
| Acetonitrile | LC50: = 1850 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 1000 mg/L, 96h static (Pimephales promelas)<br>LC50: 1600 - 1690 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 1650 mg/L, 96h static (Poecilia reticulata) |            |                  |

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| Component    | Microtox                                                               | M-Factor |
|--------------|------------------------------------------------------------------------|----------|
| Acetonitrile | EC50 = 28000 mg/L 48 h<br>EC50 = 73 mg/L 24 h<br>EC50 = 7500 mg/L 15 h |          |

**12.2. Persistence and degradability** No information available  
**Persistence** Persistence is unlikely, based on information available.

**12.3. Bioaccumulative potential** Bioaccumulation is unlikely

| Component    | log Pow | Bioconcentration factor (BCF) |
|--------------|---------|-------------------------------|
| Acetonitrile | -0.34   | No data available             |

**12.4. Mobility in soil** The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

**12.5. Results of PBT and vPvB assessment** Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

**12.6. Endocrine disrupting properties**  
**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors.

**12.7. Other adverse effects**  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**European Waste Catalogue (EWC)** According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

**Switzerland - Waste Ordinance** Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

**14.1. UN number** UN1993  
**14.2. UN proper shipping name** Flammable liquid, n.o.s.  
**Technical Shipping Name** Acetonitrile, Tetrazole

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**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** II

## ADR

**14.1. UN number** UN1993  
**14.2. UN proper shipping name** Flammable liquid, n.o.s.  
**Technical Shipping Name** Acetonitrile, Tetrazole  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** II

## IATA

**14.1. UN number** UN1993  
**14.2. UN proper shipping name** Flammable liquid, n.o.s.  
**Technical Shipping Name** Acetonitrile, Tetrazole  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** II  
**14.5. Environmental hazards** No hazards identified  
**14.6. Special precautions for user** No special precautions required.  
**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component    | CAS No   | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|--------------|----------|-----------|--------|-----|-------|------|----------|------|------|
| Acetonitrile | 75-05-8  | 200-835-2 | -      | -   | X     | X    | KE-00067 | X    | X    |
| 1H-Tetrazole | 288-94-8 | 206-023-4 | -      | -   | -     | X    | KE-33715 | -    | X    |

| Component    | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|--------------|----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Acetonitrile | 75-05-8  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| 1H-Tetrazole | 288-94-8 | -    | -                                             | -   | -    | -    | X     | -     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component    | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Acetonitrile | 75-05-8  | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |
| 1H-Tetrazole | 288-94-8 | -                                                                   | -                                                                             | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

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## Seveso III Directive (2012/18/EC)

| Component    | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Acetonitrile | 75-05-8  | Not applicable                                                                            | Not applicable                                                                           |
| 1H-Tetrazole | 288-94-8 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 2 (self classification)

| Component    | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|--------------|---------------------------------------|-------------------------|
| Acetonitrile | WGK2                                  |                         |

| Component    | France - INRS (Tables of occupational diseases)      |
|--------------|------------------------------------------------------|
| Acetonitrile | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed

H312 - Harmful in contact with skin

H332 - Harmful if inhaled

H319 - Causes serious eye irritation

H201 - Explosive; mass explosion hazard

H225 - Highly flammable liquid and vapor

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

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KECL - Korean Existing and Evaluated Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

**Creation Date** 20-Jul-2010

**Revision Date** 09-Feb-2024

**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**